

International Economic Development

ECON 4660/8666

Lecture 9 Fiscal Policy

- Government plays an important role in providing public goods, reallocating resources, redistributing income, regulating markets, and maintaining macroeconomic stability and good business environment.
- Government is involved in education, healthcare, infrastructure, national defense, social safety net provisions, capital market regulations, environmental protection, etc.
- Government collects taxes and spends the tax revenues.
- Fiscal policies:
Changes in government spending and taxes

I. Government Expenditures

- Recurrent Expenditures (government consumption)
 - recur year after year, e.g., wages, interest payment, health, education, subsidies, transfers
- Capital Expenditures
 - investment, on bridges, highways.

A. Wages and salaries of civil servants, teachers, and the military personnel

- Large fraction of government expenditures in the developing countries

- The case of Mali:

Population: 11 million in 2002

per capita income: \$240 in 2002; Gross National Income.

Life expectancy at birth: 48 years 2000-2005

38 years 1970-1975

- After independence in 1960, civil service expanded rapidly. In 1983, total public sector accounted for 63% of the employment.
- Reforms: Competitive exams for entry level positions,
voluntary early departure program,
hiring ceiling

B. Government Purchases.

- Goods: Computers, tables, equipment and materials for employees, military equipment.
- Services: Construction, janitorial work,

C. Interest payments

- Interests on government debt.
- 7.6% total expenditures in US in 2022.

D. Subsidies.

- Subsidies to consumers, social welfare payment, subsidies to state enterprises.
- 63% total expenditures in Brazil in 1998.

E. Intergovernmental transfers.

- 30% total expenditure in Japan in 2001.
- Central government, 65% rev. 35% exp.

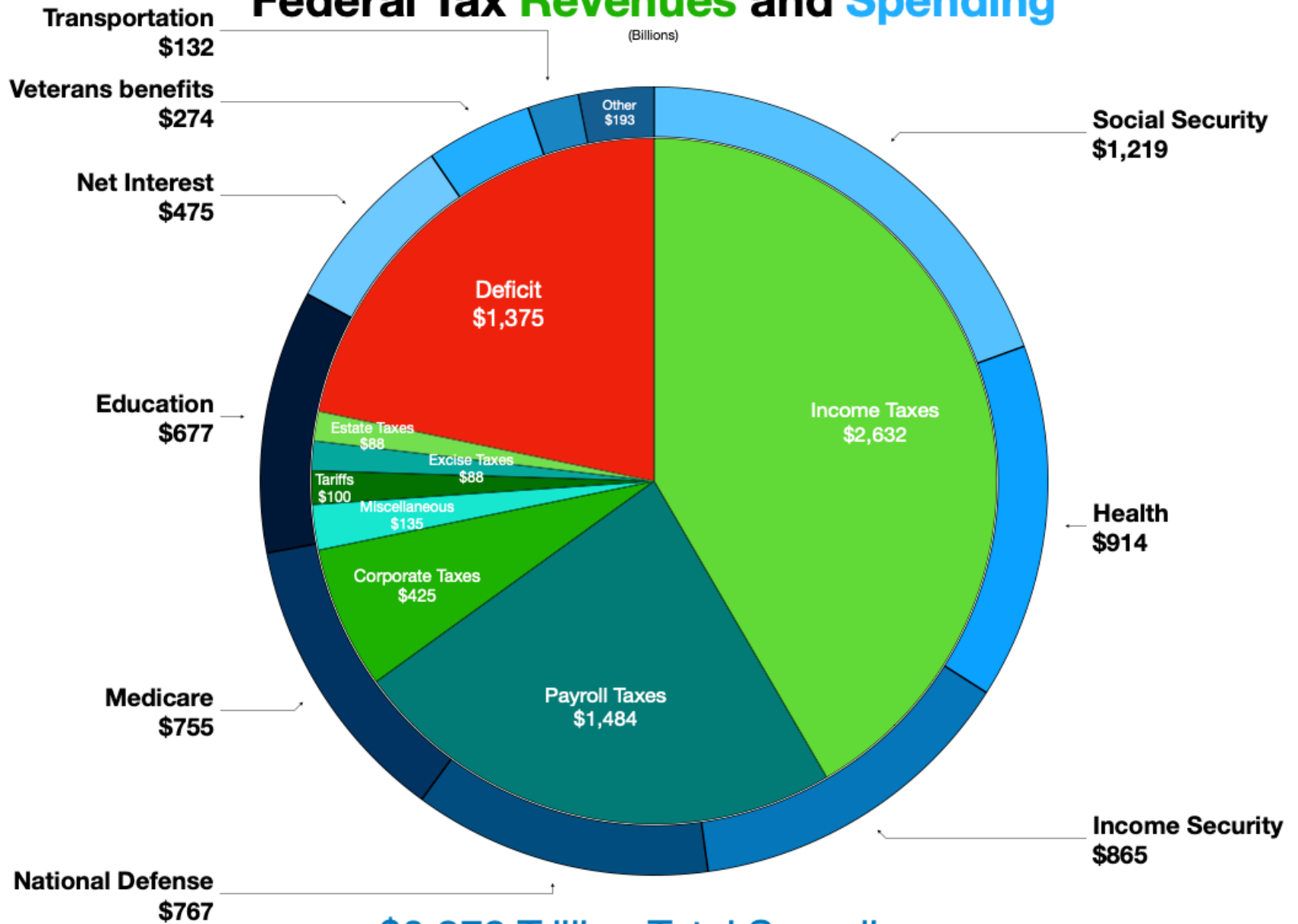
General Government Outlays by Function for Countries
with Data Available for 2006-2007 (% of Total Outlays)

Countries	Year	General Public Services	Defense	Public Order & Safety	Econo- mic Affairs	Environ- mental Protec- tion	Com- munity Ameniti- es	Health	Recrea- tion, Culture, & Religion	Edu- cation
France	2007	13.28	3.39	2.41	5.36	1.66	3.62	13.72	2.90	11.24
Germany	2007	13.61	2.41	3.51	7.23	1.10	1.93	14.01	1.36	9.09
Italy	2007	15.98	3.14	3.89	10.23	1.83	2.16	12.57	3.05	11.22
Luxembourg	2007	15.74	0.61	2.52	10.80	2.65	2.05	11.90	5.39	13.07
Netherlands	2007	16.11	3.01	3.95	10.48	1.81	2.06	12.54	2.95	11.17
Slovenia	2006	11.38	3.09	4.18	10.68	1.34	1.30	13.66	2.83	14.95
Spain	2007	11.47	2.64	4.86	12.56	2.36	2.45	14.70	4.09	11.28
Others										
Australia	2007	11.43	4.85	5.02	12.41	1.61	2.82	18.02	2.43	14.27
Canada	2007	12.52	2.86	4.95	9.09	1.81	2.04	19.01	2.62	15.02
Hong Kong	2006	24.40	—	10.00	7.47	3.11	6.22	13.23	3.40	19.32
Denmark	2007	12.54	3.08	1.98	6.63	1.11	1.34	14.12	3.07	14.50
Iceland	2007	13.13	0.15	3.46	13.73	1.49	1.15	18.67	8.90	19.18
Israel	2007	14.43	17.07	3.88	6.09	1.49	1.23	11.32	3.09	16.14
Japan	2006	12.91	2.59	3.89	10.55	3.55	1.81	19.60	0.43	10.74
New Zealand	2007	12.58	2.98	4.66	9.58	1.18	1.94	15.31	4.33	15.39
Norway	2007	10.16	3.97	2.23	9.03	1.35	1.44	17.42	2.61	13.31
San Marino	2002	14.52	.	2.94	14.32	3.23	2.34	18.14	2.30	9.40
Singapore	2007	12.37	27.99	6.24	9.81	.	12.20	6.04	0.48	20.82
Switzerland	2006	13.78	2.89	5.01	11.96	1.71	1.12	12.17	2.62	16.86
US	2007	13.47	11.54	5.71	9.98	.	1.85	21.06	0.87	16.93

Countries	Year	General Public Services	Defense	Public Order & Safety	Economic Affairs	Environmental Protection	Community Amenities	Health	Recreation, Culture, & Religion	Education
WAEMU										
Mauritius	2007	22.25	0.65	7.78	10.70	5.18	2.90	8.84	1.53	15.08
Seychelles	2005	39.72	3.35	4.28	5.42	.	.	.	.	0.80
Asia										
Bhutan	2000	22.40	.	5.44	46.38	.	.	.	.	2.27
China, Mainland	2006	18.20	5.24	4.98	37.68	3.21	0.44	2.51	1.27	.
Macao	2007	18.32	.	16.60	13.88	5.79	1.01	10.54	6.51	
Maldives	2007	21.89	5.04	7.24	8.32	1.13	24.90	8.12	4.10	14.22
Thailand	2007	5.77	.	.	.	.	.	.	.	.
Central and Eastern Europe										
Albania	2006	21.57	4.42	6.26	15.23	.	5.65	8.24	1.27	10.77
Bulgaria	2007	12.55	6.01	8.12	12.50	0.67	3.17	12.72	2.26	10.19
Croatia	2007	11.21	3.55	6.07	13.99	0.66	4.71	13.84	3.00	9.69
Czech Republic	2007	11.94	3.26	5.22	17.01	2.65	3.57	14.24	2.61	9.65
Latvia	2007	11.48	4.03	7.06	17.23	2.72	3.53	10.09	4.60	16.32
Lithuania.	2007	9.67	5.21	4.73	11.65	2.50	1.45	15.26	3.23	15.26
Poland	2007	12.58	3.89	4.72	10.15	1.40	1.70	10.62	2.64	12.72
Romania	2006	12.61	3.56	7.07	19.36	0.98	3.96	9.48	2.63	12.09
Slovak Republic	2007	14.15	3.97	5.50	13.03	1.95	2.03	17.86	2.99	9.55

Federal Tax Revenues and Spending

(Billions)

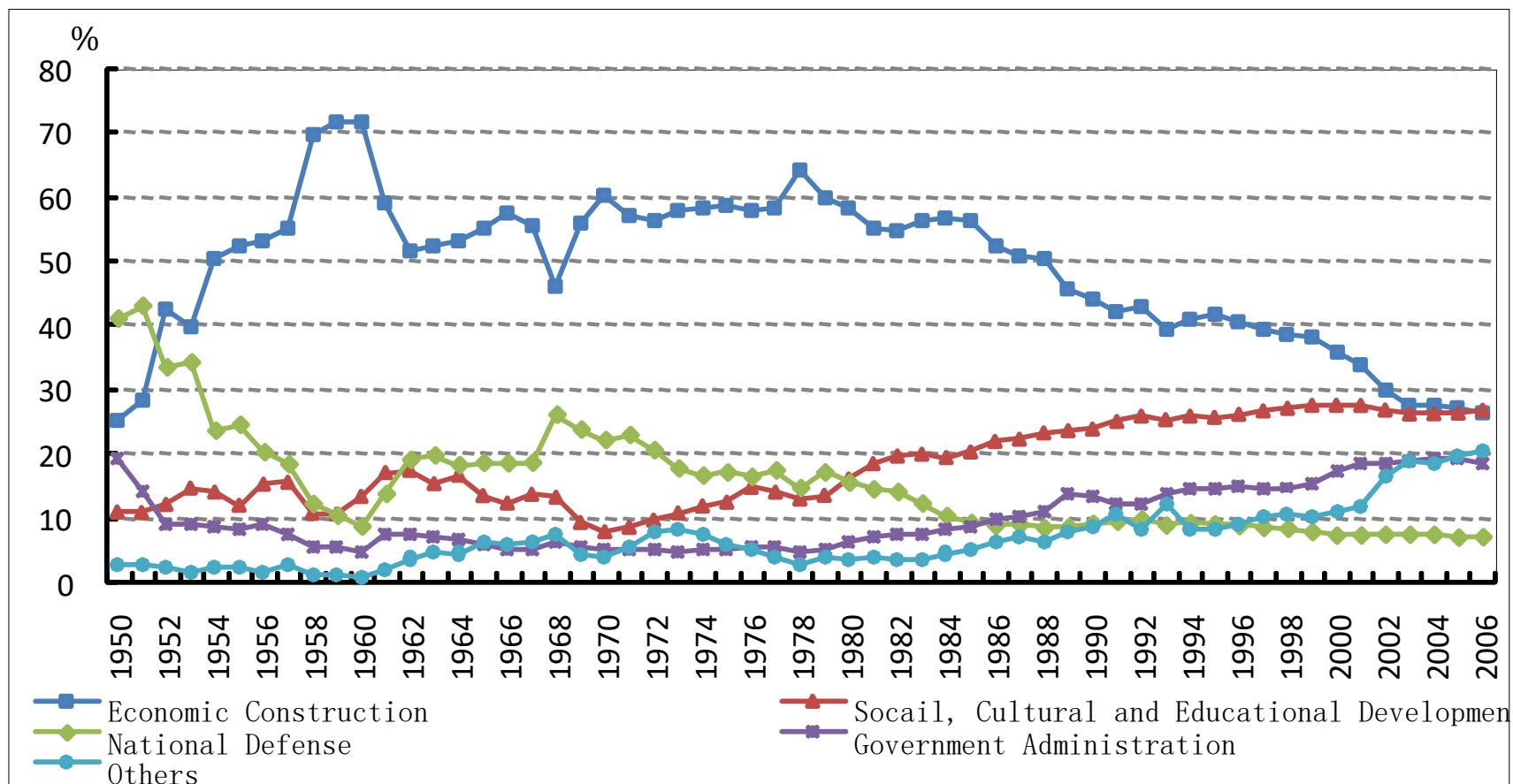


\$6.272 Trillion Total Spending

\$4.896 Trillion Total Revenue

\$1.375 Trillion Total Deficit

Government Budgetary Expenditures by Function 1950-2006



II. Taxes

- Goals of taxation:
 - Efficient resource allocation,
 - A satisfactory distribution of income,
 - Full employment with price stability,
 - A high stable growth rate of output
- General Criteria to evaluate different taxes:
 - Allocational efficiency
 - Equity
 - Administrative feasibility
 - Revenue productivity
- Taxes put resources in the hands of government, to carry certain investment programs good for the economy in the long run.

Raising tax revenues is always difficult in LDCs.

- underdeveloped infrastructure, rudimentary accounting systems, and prevalent tax evasion.

1. Taxes on International Trade

Tariffs: taxes on imports

In some African LDCs, Gambia, Uganda Sudan, import taxes once account for half of government revenues.

- Tariffs → Smuggling
 - China, Cars
 - T.V.
 - Camera
- Export taxes on raw materials.
 - Example: Colombia, coffee,
Gambia, peanuts.
Sri Lanka, tea
- Export taxes unconstitutional in the U.S.

2. Personal and Corporate Income Taxes

(Direct taxes)

- Few LDCs can rely heavily on personal income tax.
 - Underreporting of Income, Tax loopholes. Tax evasion.
- Corporate income taxes also not effective in raising revenues.
 - Size of the Corporate Sector, transfer pricing, tax evasion

3. Sales and Excise Taxes (Indirect taxes), are more effective in raising tax revenues.

- Sales taxes: taxes on all products except food, medicine, etc.
- Excise taxes: Taxes on specified goods,
e.g. tobacco, alcoholic beverages, gas.
- Both sales and excise taxes are consumption-based, meaning they are applied when a consumer purchases a product or service.
 - Their effectiveness in raising revenue often stems from their broad base and the fact that they can be harder to evade than income-based taxes.
 - Additionally, they can be used as a policy tool to influence consumer behavior, such as discouraging the consumption of tobacco or alcohol through higher tax rates.

4. Value-added taxes: indirect taxes that are imposed on the value added to goods and services at each stage of production or distribution.

- Two types of value-added taxes:
 - A general consumption tax. Permits firms to subtract purchases of capital goods in computing the tax base. Total value-added is thus total retail sales of final consumer goods.
 - A tax on national income. Purchase of capital goods are not deducted, instead, firms are permitted to deduct an allowance for depreciation over useful life of the asset.
 - In practice, the choice between these two methods can influence the behavior of firms and individuals. For example, a general consumption tax might encourage savings and investments since it doesn't tax them, whereas a tax on national income could be seen as a broader approach to taxation, capturing more aspects of economic activity.

**Composition of Taxes and Other Government Revenues
for Countries with Data for 2008-20089(%)**

Country or Region	Year	Taxes							Social Contributions as % of GDP	Grants as % of GDP	Other Revenue as % of GDP
		Income, Profits, Capital Gains		Property		Goods and Services		Total Taxes as % of GDP			
		As % of GDP	As % of Total Tax Revenue	As % of GDP	As % of Total Tax Revenue	As % of GDP	As % of Total Tax Revenue				
United States	2009	9.61	55.74	3.24	18.79	4.23	24.54	17.24	6.91	-	6.67
Canada	2009	14.29	54.65	3.55	13.58	6.94	26.54	26.15	4.82	-	6.95
Australia	2009	16.03p	59.44	2.22p	8.23	6.89p	25.55	26.97p	-p	-p	6.48p
Japan	2009	7.26	44.90	0.28	1.73	2.58	15.96	16.17	11.71	-	4.94
New Zealand	2007	21.70p	62.68	1.73p	5.00	10.12p	29.23	34.62p	0.06p	-p	6.66p
Austria	2009	12.14p	44.02	0.55p	1.99	12.21p	44.27	27.58p	16.64p	0.12p	4.46p
Belgium	2009	14.66p	51.57	...		...		28.43p	16.81p	...	...
Finland	2009	16.86p	54.72	1.11p	3.60	12.84p	41.67	30.81p	12.20p	0.21p	10.16p
France	2009	8.75	35.25	4.48	18.05	10.51	42.34	24.82	18.45	...	...
Germany	2009	11.91	50.13	0.79	3.32	11.05	46.51	23.76	17.11	0.14	3.50
Greece	2009	7.69	39.04	1.03	5.23	10.78	54.72	19.70	13.21	0.22	5.02
Ireland	2009	10.33	46.62	1.04	4.69	...		22.16	7.20	...	...
Italy	2009	14.18	48.80	1.45	4.99	11.54	39.71	29.06	14.14	0.20	3.24
Luxembourg	2009	13.17	51.09	1.30	5.04	11.30	43.83	25.78	12.07	0.22	3.38
Netherlands	2009	10.75	44.77	1.32	5.50	11.85	49.35	24.01	14.58	0.05	7.33
Portugal	2009	8.64p	39.69	1.01p	4.64	11.28p	51.81	21.77p	11.93p	...	...
Spain	2009	9.34p	50.03	2.07p	11.09	7.24p	38.78	18.67p	13.31p	0.55p	2.14p
Denmark	2009	29.08p	61.75	2.67p	5.67	14.66p	31.13	47.09p	1.94p	...	...
Iceland	2009	15.98	52.14	2.08	6.79	11.65	38.01	30.65	3.06	0.14	7.08
Norway	2009	18.65p	59.13	1.20p	3.80	11.57p	36.68	31.54p	9.86p	0.01p	14.59p
Sweden	2009	19.26	50.27	1.05	2.74	13.45	35.11	38.31	8.68	0.07	6.98
Switzerland	2008	†13.82	61.48	†2.16	9.61	†5.36	23.84	†22.48	†6.69	†0.02	†5.13
UK	2009	13.24	47.64	4.38	15.76	10.10	36.34	27.79	8.56	0.42	3.58

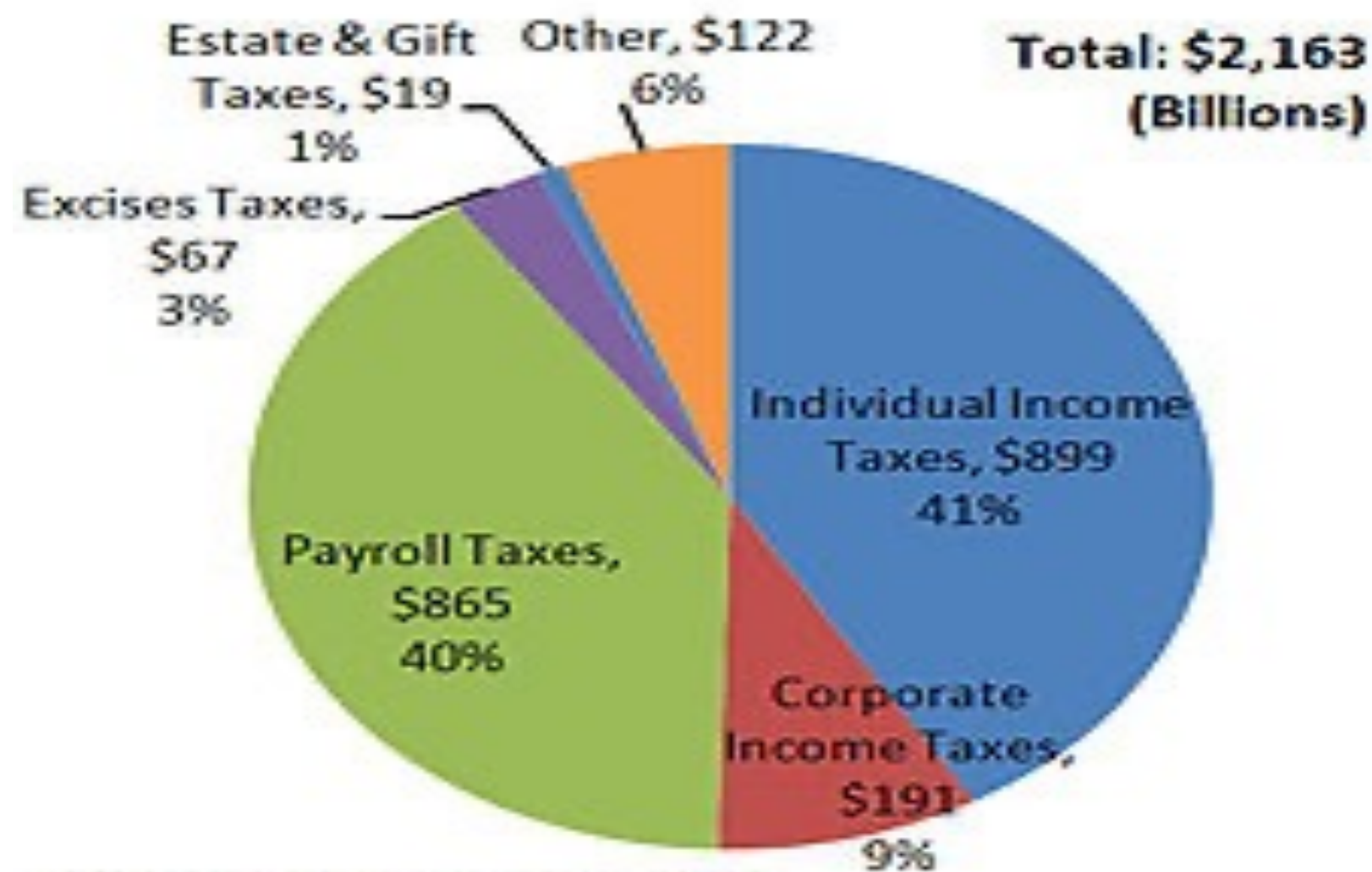
Country or Region	Year	Taxes							Social Contributions as % of GDP	Grants as % of GDP	Other Revenue as % of GDP
		Income, Profits, Capital Gains		Property		Goods and Services		Total Taxes as % of GDP			
		As % of GDP	As % of Total Tax Revenue	As % of GDP	As % of Total Tax Revenue	As % of GDP	As % of Total Tax Revenue				
Lesotho	2007	10.94	18.96	-		9.43	16.35	57.69	-	1.45	6.53
Mauritius	2009	†5.47	28.62	†1.53	8.01	†11.26	58.92	†19.11	†0.99	†1.00	†3.29
Seychelles	2005	7.28	20.26	-		19.57	54.47	35.93	11.24	1.35	9.55
South Africa	2009	14.70p	55.68	1.45p	5.49	9.18p	34.77	26.40p	0.63p	0.07p	7.29p
Hong Kong	2008	8.83	66.94	2.45	18.57	1.73	13.12	13.19	-	-	6.87
Macao	2009	1.66p	5.44	0.46p	1.51	28.23p	92.53	30.51p	0.10p	-p	3.62p
Maldives	2009	1.46p	9.29	0.02p	0.13	3.53p	22.46	15.72p	-p	3.57p	15.94p
Singapore	2009	6.42	46.93	0.74	5.41	4.72	34.50	13.68	-	-	4.33
Thailand	2009	7.21p	43.07	0.32p	1.91	8.19p	48.92	16.74p	0.93p	0.02p	3.11p
Albania	2008	4.15	21.04	0.15	0.76	14.22	72.11	19.72	4.34	0.39	2.31
Armenia	2009	4.58	26.50	0.49	2.84	9.52	55.09	17.28	3.32	0.75	2.67
Belarus	2009	7.04	23.31	1.19	3.94	14.74	48.81	30.20	11.55	-	5.72
Bulgaria	2009	5.56	24.85	1.39	6.21	15.23	68.08	22.37	7.69	1.71	4.23
Croatia	2009	5.95	26.94	0.31	1.40	14.91	67.50	22.09	12.01	0.19	4.17
Cyprus	2009	10.43p	39.70	0.57p	2.17	13.79p	52.49	26.27p	8.64p	...	...
Czech	2009	7.02p	38.55	0.40p	2.20	10.79p	59.25	18.21p	14.06p	2.20p	2.91p
Estonia	2009	7.54	33.63	0.35	1.56	14.53	64.81	22.42	13.29	...	...
Georgia	2009	9.12	37.30	0.89	3.64	13.90	56.85	24.45	-	2.17	2.71
Hungary	2009	9.50	36.29	0.84	3.21	15.50	59.21	26.18	13.10	1.28	5.55
Kazakhstan	2009	5.99	43.69	0.79	5.76	3.90	28.45	13.71	-	7.10	1.47
Kyrgyz	2009	...		...		...		...	...	...	...
Latvia	2009	7.04	39.31	0.56	3.13	10.14	56.62	17.91	8.92	4.04	4.55
Lithuania	2009	5.98p	34.23	0.36p	2.06	11.11p	63.59	17.47p	12.84p	2.16p	2.40p
Malta	2009	13.02p	46.78	1.08p	3.88	13.74p	49.37	27.83p	7.46p	0.99p	3.21p
Moldova	2009	3.20	14.59	0.43	1.96	16.77	76.47	21.93	11.71	1.91	4.23

Country or Region	Year	Taxes							Social Contributions as % of GDP	Grants as % of GDP	Other Revenue as % of GDP
		Income, Profits, Capital Gains		Property		Goods and Services		Total Taxes as % of GDP			
		As % of GDP	As % of Total Tax Revenue	As % of GDP	As % of Total Tax Revenue	As % of GDP	As % of Total Tax Revenue				
Poland	2009	6.93	33.97	1.12	5.49	11.77	57.70	20.40	11.32	0.95	4.33
Romania	2008	6.62	36.04	0.63	3.43	10.99	59.83	18.37	10.10	0.42	3.26
Russia	2009	7.80	34.98	1.29	5.78	6.61	29.64	22.30	5.98	-	23.28
Slovak	2009	†0.18p	34.62	†0.01p	1.92	†0.34p	65.38	†0.52p	†0.42p	†0.03p	†0.14p
Slovenia	2009	7.93	36.54	0.59	2.72	13.02	60.00	21.70	15.93	1.54	4.66
Ukraine	2009	8.68	38.87	0.31	1.39	12.54	56.16	22.33	12.99	0.07	6.69
Israel	2009	9.58	37.50	2.87	11.23	11.51	45.05	25.55	6.54	1.06	6.35
Kuwait	2008	...		...		...		...	...	...	...
Argentina	2004	1.29p	22.55	0.66p	11.54	2.74p	47.90	5.72p	0.75p	-p	0.87p
Bolivia	2007	3.08	12.56	2.89	11.79	16.68	68.03	24.52	1.83	2.65	7.81
Chile	2009	5.64	33.79	0.57	3.42	9.98	59.80	16.69	1.50	-	3.50
Costa Rica	2009	...		...		...		...	...	...	...
El Salvador	2009	4.73	36.30	0.06	0.46	7.36	56.49	13.03	2.20	0.17	3.28
Jamaica	2009	...		...		...		...	...	...	...
Paraguay	2009	3.10	22.64	0.35	2.56	8.64	63.11	13.69	4.08	0.36	6.23
Peru	2009	5.33	37.80	0.25	1.77	6.93	49.15	14.10	1.74	0.10	2.83

Source: International Monetary Fund, Government Finance Statistics Yearbook, 2010.

Federal Receipts by Source, 2010

**Total: \$2,163
(Billions)**



Source: Joint Committee on Taxation

Total Direct Revenue US from FY 1902 to FY 2015

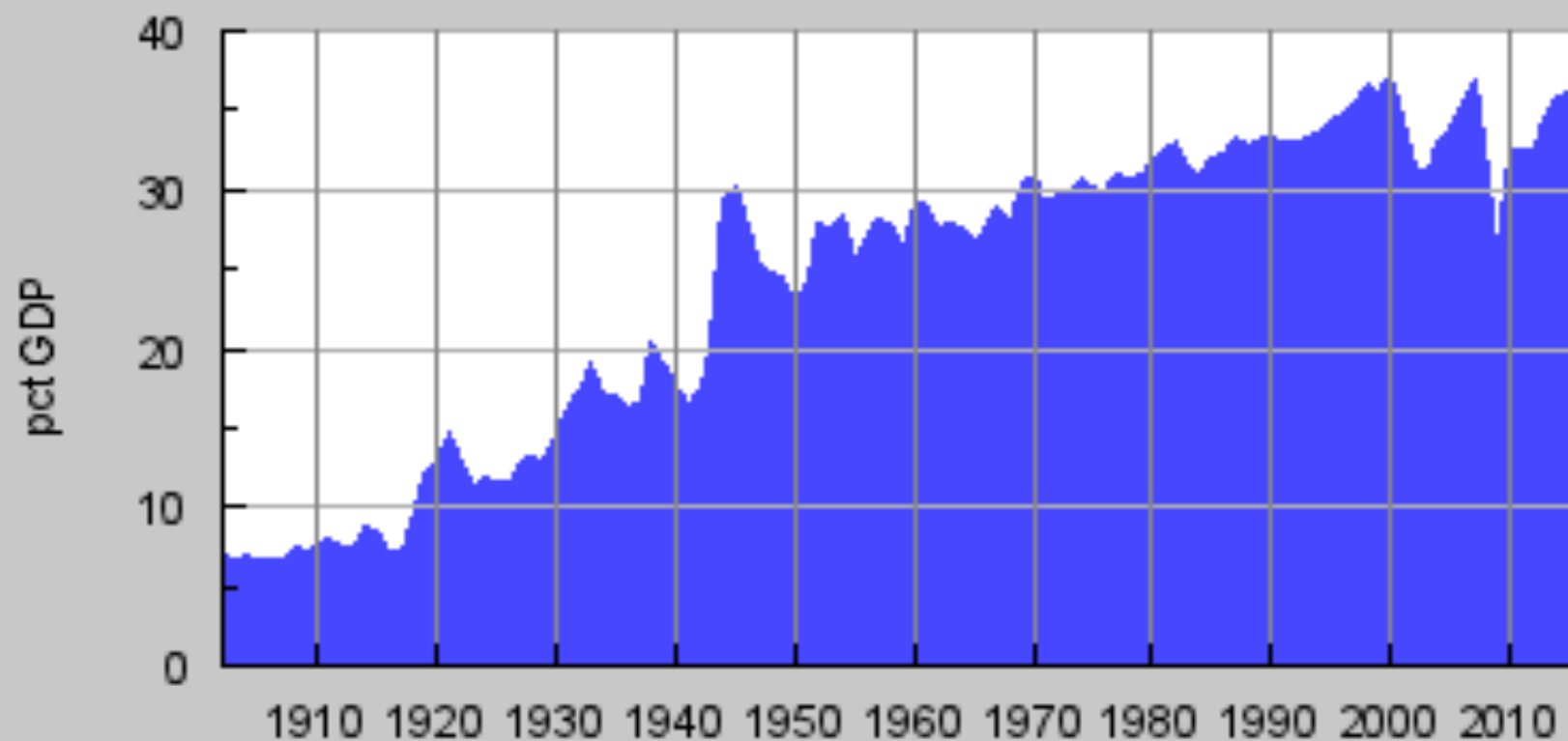
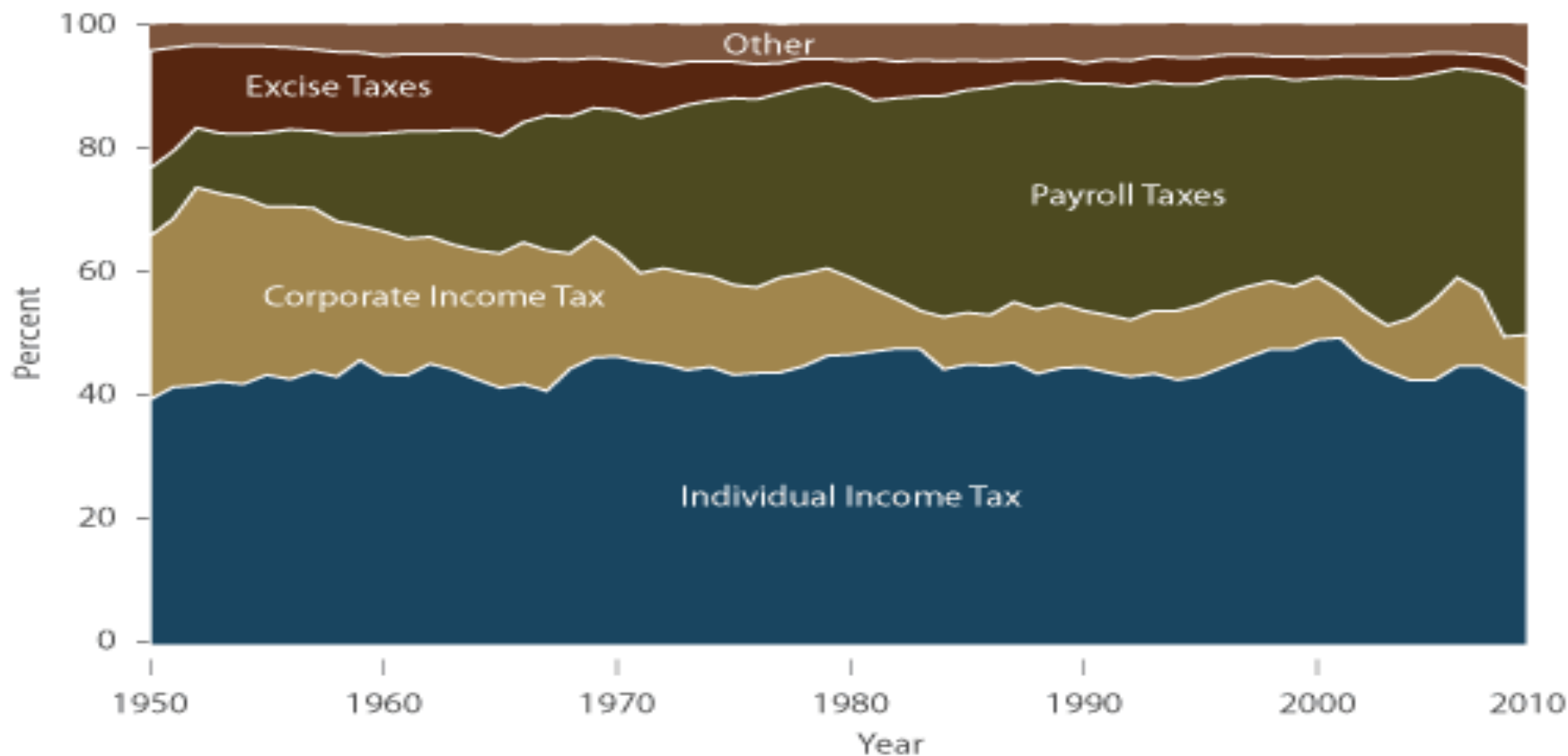


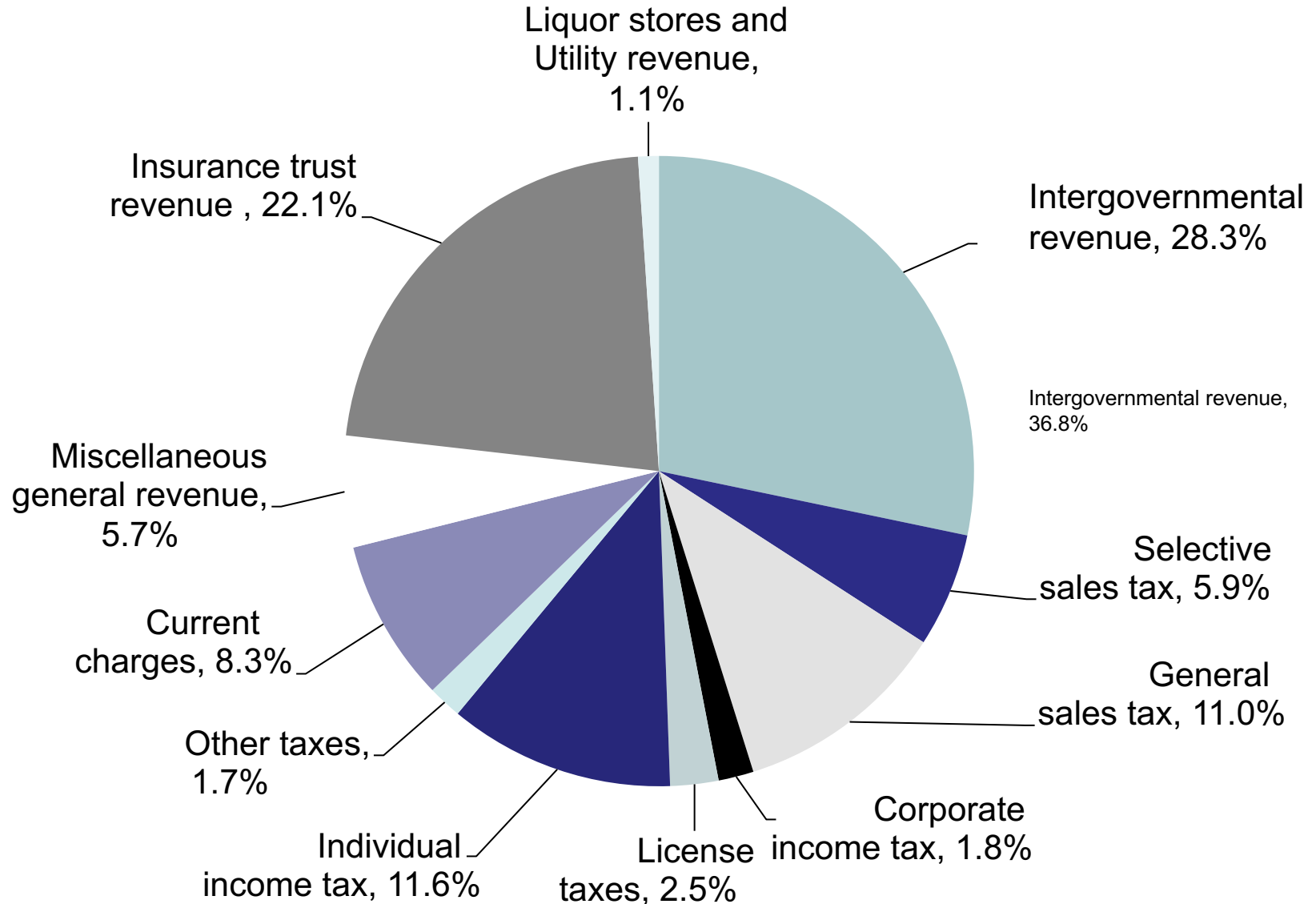
Figure 2. Sources of Federal Revenue, Fiscal 1950-2010



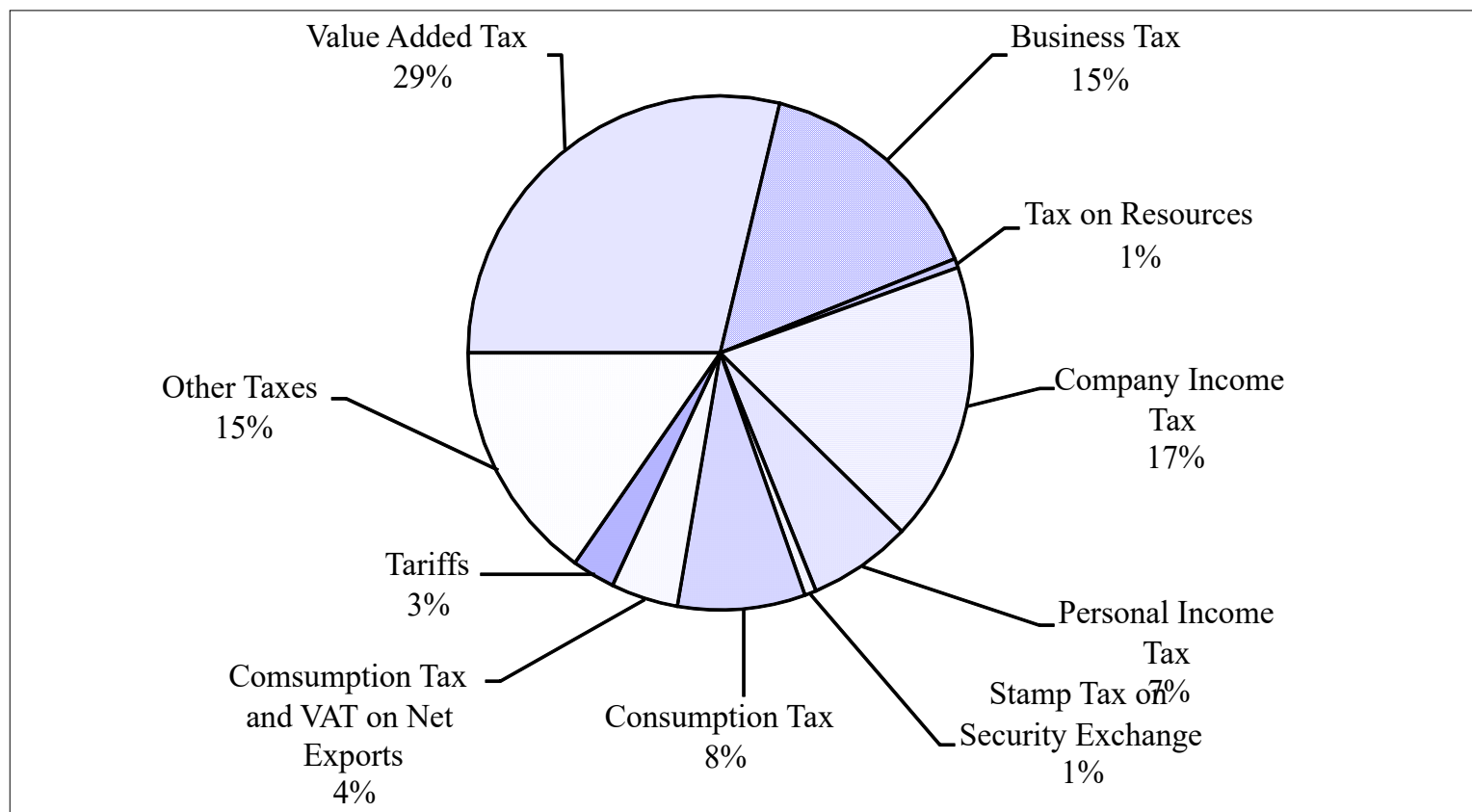
Source: Budget of the United States Government, Fiscal Year 2012, Historical Tables: Table 2.1;
<http://www.whitehouse.gov/omb/budget/Historicals>

Year	Maximum taxable earnings	OASDI tax rate	HI tax rate
1937	3,000	2%	-
1950	3,000	3%	-
1960	4,800	6%	-
1965	4,800	7.25%	-
1966	6,600	7.7%	0.7%
1970	7,800	8.4%	1.2%
1971	7,800	9.2%	1.2%
1975	14,100	9.9%	1.8%
1980	25,900	10.16%	2.1%
1985	39,600	11.4%	2.7%
1990	51,300	12.4%	2.9%
1995	61,200	12.4%	2.9%
2000	76,200	12.4%	2.9%
2005	90,000	12.4%	2.9%
2006	94,200	12.4%	2.9%
2007	97,500	12.4%	2.9%
2008	102,000	12.4%	2.9%
2009	106,800	12.4%	2.9%
2010	106,800	12.4%	2.9%
2011	106,800	10.4%	2.9%

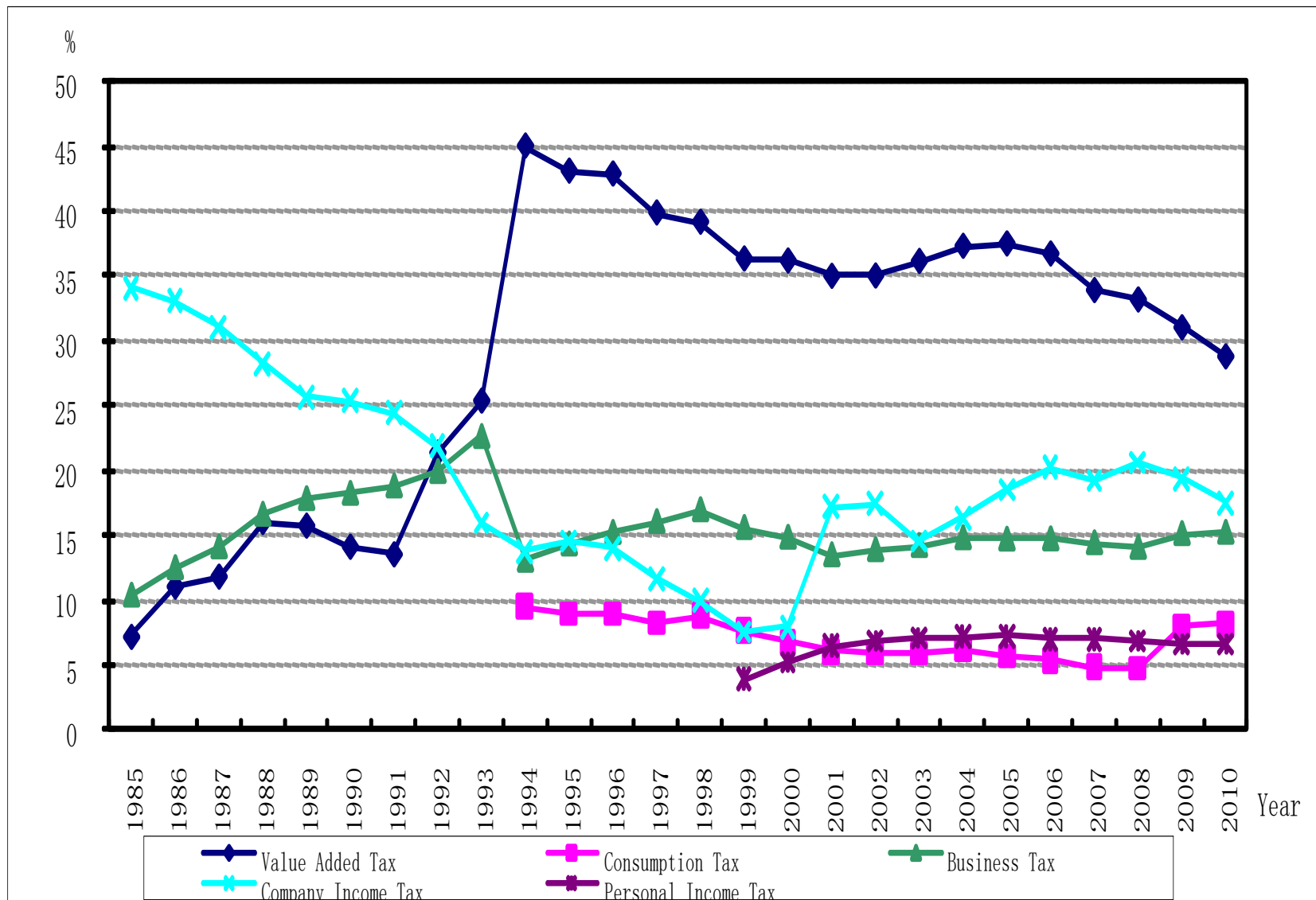
Sources of State Government Revenue



Composition of China's Tax Revenue in 2010



Share of China's Major Taxes in Total Tax Revenue



Tax Reforms:

- Widening the Tax Base:

- Advantages:

- It reduces the tax burden on any single group or individual.
 - It can lead to more stable and predictable revenue streams for the government.
 - Helps in reducing tax evasion as more entities are brought into the tax net.

- Disadvantages:

- It might be seen as regressive, as lower-income individuals might end up paying a higher proportion of their income in taxes.
 - Implementation can be complex, especially if the tax system was previously narrow.

Tax Reforms:

- Lower Tax Rate and Improve Tax Administration:
 - Advantages:
 - Lower tax rates can incentivize businesses to invest and expand, potentially leading to economic growth.
 - Improved administration can reduce bureaucracy, making it easier for individuals and firms to comply.
 - It can also reduce tax evasion and increase government revenue.
 - Disadvantages:
 - If not done correctly, lowering tax rates can lead to reduced government revenue, affecting public services.
 - There's a risk of favoring certain groups or sectors over others.

Tax Reforms:

- Adoption of Value-Added Tax (VAT):
 - Advantages:
 - VAT is considered more efficient than a sales tax as it reduces the cascading effect of tax, i.e., tax on tax.
 - It promotes transparency in the taxation system.
 - As a consumption tax, it can encourage savings and investments.
 - Disadvantages:
 - VAT can be seen as regressive since everyone pays the same rate regardless of their income.
 - Implementation can be complex, especially if transitioning from a different system.

Tax Reforms:

- Transition from Income Tax to Consumption Tax:
 - Advantages:
 - Encourages savings and investments as consumption taxes don't tax income that is saved.
 - Can lead to increased capital accumulation.
 - Disadvantages:
 - Consumption taxes can be regressive, placing a heavier burden on lower-income individuals.
 - Might discourage consumer spending, which can slow down the economy.

Bolivia

- Economic Challenges in Bolivia in the Mid-1980s:
 - Hyperinflation: One of the most significant challenges was hyperinflation, which reached a staggering 24,000% in 1985. Such high inflation rates eroded purchasing power, causing widespread poverty and reducing the real income of Bolivians.
 - Unemployment: The unemployment rate was at a high of 17%, further straining the socio-economic fabric of the country.
 - Currency Substitution: Given the extreme devaluation of the Bolivian peso and the lack of confidence in the national currency, Bolivians started using the US dollar as a primary store of value and medium of exchange. This phenomenon, known as "Currency Substitution" or "Dollarization," reflected the public's lack of trust in the country's economic policies.
 - Low Tax Revenues: Tax revenues were extremely low, at just 1% of GDP. This figure highlighted the government's inability to collect taxes and the large informal economy that bypassed official channels.

Bolivia

- Reforms in 1986: To counter these challenges and stabilize the economy, the Bolivian government initiated reforms in 1986. Some of the key measures included:
 - Fiscal Discipline: The government implemented stringent controls on spending. One of the most significant measures was to authorize payments only to the extent of taxes collected each day. This move was aimed at living within means and ensuring fiscal discipline.
 - Monetary Reforms: To restore confidence in the national currency, the government undertook monetary reforms, including introducing a new currency.
 - Tax Reforms: Efforts were made to broaden the tax base, improve tax collection, and streamline the tax system.
 - Structural Reforms: These reforms included privatization of state-owned enterprises, liberalization of trade and investment, and regulatory reforms to encourage private sector participation and growth.

Taxes and Economic Development

- Taxes and private Investment
 - Public vs. Private Savings:
 - When taxes are increased, the government can accumulate more public savings, which can be used to fund public projects and services.
 - However, higher taxes can decrease private savings as individuals have less disposable income.
 - Investment Decisions:
 - High taxes can deter private investment as they reduce potential returns on investments.
 - Conversely, tax incentives or breaks can encourage private sector investments in specific industries or regions.

Taxes and Economic Development

- Taxes and Capital Mobility
 - Attracting Foreign Capital:
 - Countries with competitive tax rates can attract foreign investments, leading to economic growth.
 - Such investments can be in the form of Foreign Direct Investment (FDI) or portfolio investments.
 - Preventing Capital Flight:
 - High taxes can result in capital flight, where domestic investors move their investments abroad to countries with more favorable tax regimes.
 - The example of Hong Kong investors moving to Canada post-1997 due to political uncertainties, and later returning because of HK's competitive 16.5% corporate income tax, underscores this point.

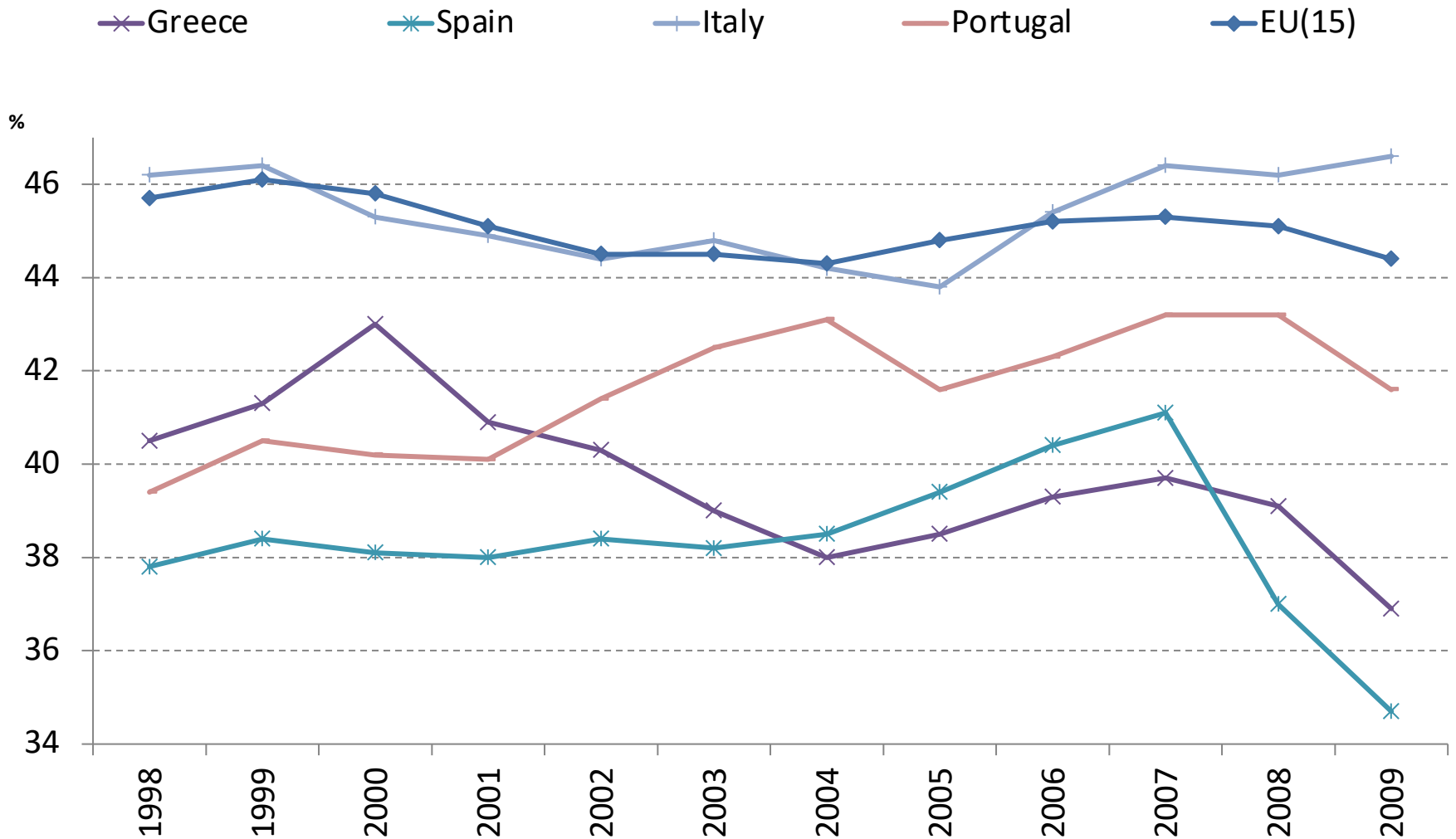
Taxes and Economic Development

- Taxes and Income Distribution
 - Ability to Pay Principle:
 - This principle suggests that taxes should be levied based on an individual's ability to pay.
 - Progressive taxation, where higher-income individuals are taxed at higher rates than those with lower incomes, is an example of this principle in action.
 - Benefit Received Principle:
 - This principle suggests that individuals should be taxed based on the benefits they receive from public services.
 - For instance, if someone benefits more from public roads because they own a transportation company, they might be expected to pay more in taxes.

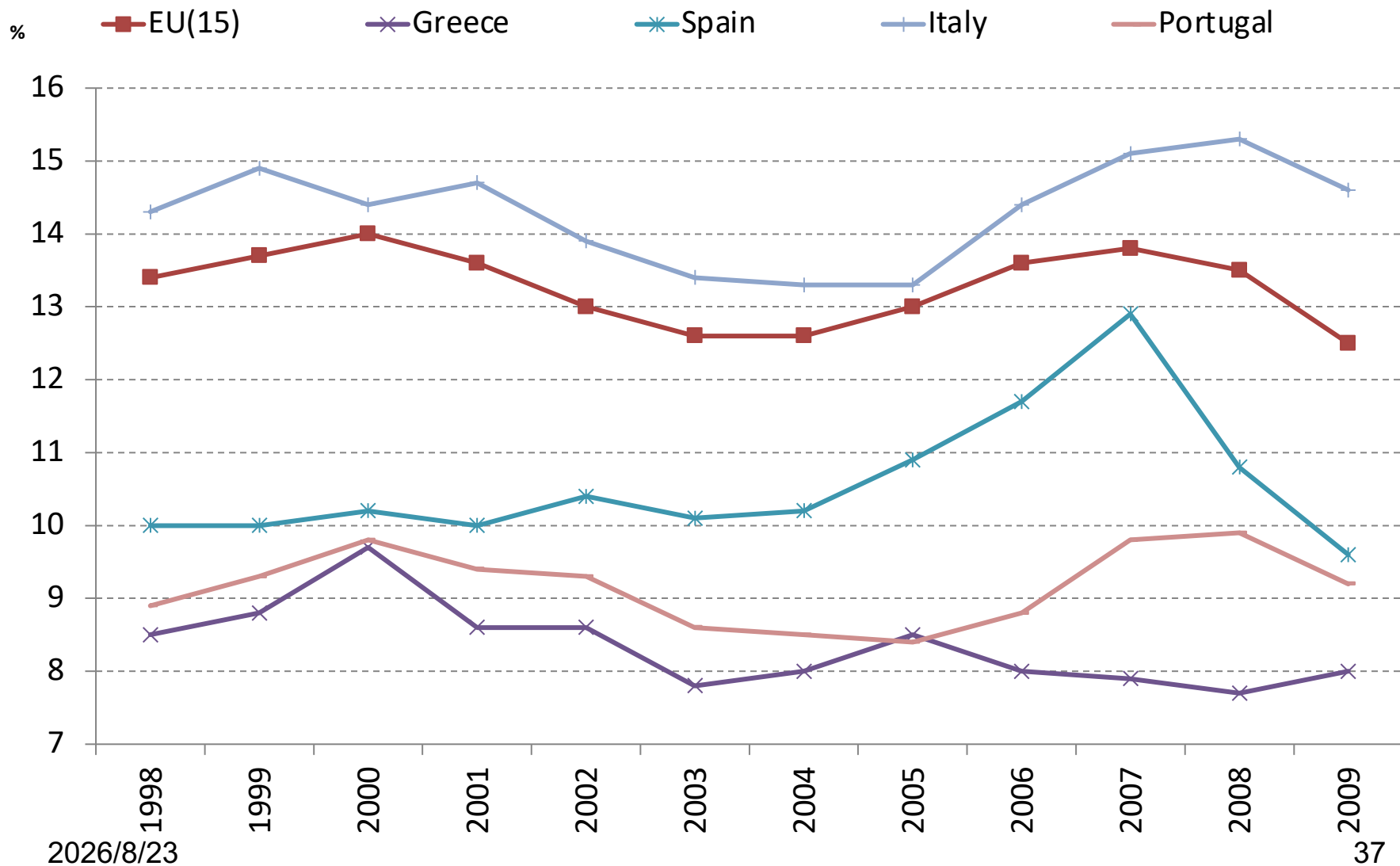
Taxes and Economic Development

- Taxes on Luxury Consumption
 - Taxes on luxury goods can serve multiple purposes.
 - Firstly, they can be a source of additional revenue for the government.
 - Secondly, such taxes can also be used to address wealth disparities by placing a higher tax burden on the affluent who consume luxury goods.
 - Additionally, taxing luxury consumption can discourage excessive consumption and promote more sustainable consumption patterns.

Tax Revenue-GDP Ratios (1998-2009)



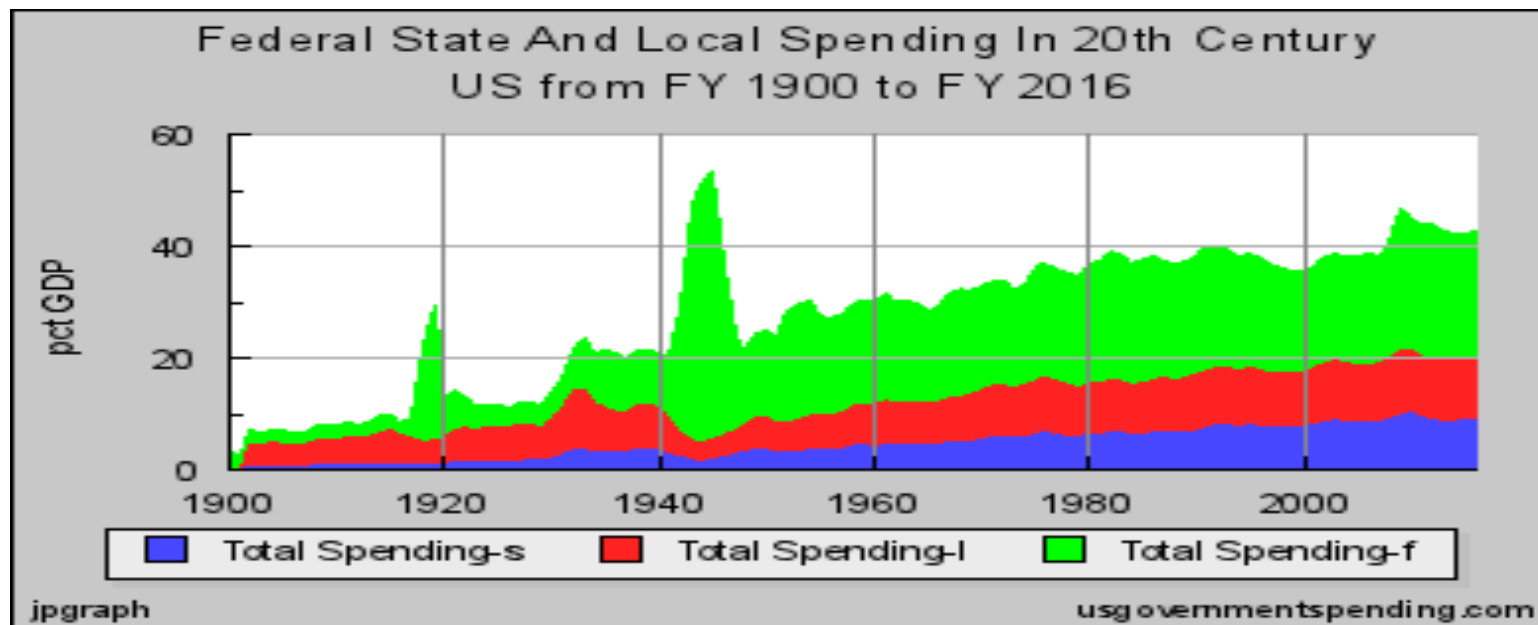
Income Tax-GDP Ratios (1998-2009)



III. Government and Economic Development

III.1 Economic Development and Government Size

- Wagner's law: the relative size of government tends to increase as per capita income increases $\frac{G}{GDP}$
 - An observation made by Adolph Wagner (1835–1917).
- The law is based on three claims:
 - i. economic growth results in an increase in complexity requiring continued introduction of new laws and development of the legal structure;
 - ii. urbanization increases negative externalities, such as congestion and crime, which necessitate intervention;
 - iii. the goods supplied by the public sector have a high income elasticity of demand. If the elasticity of demand exceeds one, public sector expenditure will consequently rise as a proportion of income.
- Empirical evidence has failed to convincingly support that the elasticity is greater than one.
- But government size has increased in many countries over time, and the rich countries seem to have a larger size of government.



Sources: http://www.usgovernmentspending.com/us_20th_century_chart.html

- At the start of the 20th century, out of a total of 7% of GDP, 4% was spent at the local level. Federal spending spiked in World War I, but in the 1920s, local government spending still accounted for half.
- In the 1930s, federal spending surged to near half.
- After the spike of World War II the federal share increased, and state share as a percent of GDP also increased.
- By the 2010s federal spending was over 20% of GDP, state 8 to 9% of GDP, and local spending exceeded 10% of GDP.

III.2 Government and Economic Growth

A. Government spending

- The share of government spending in GDP is negatively related to per capital GDP growth.
 - Landau, D. (1983), Government Expenditure and Economic Growth: A Cross-Country Study, Southern Economic Journal, 49, pp.783-92.
 - Landau, D. (1986), Government and Economic Growth in the Less Developed Countries: An Empirical Study for 1960-1980, Economic Development and Cultural Change, 1986
 - Barro, R. (1991), “Economic Growth in a Cross Section of Countries.” Quarterly Journal of Economics, 106, May, 407-443.
- The growth rate of govt spending is positively related to economic growth.
 - Ram, Rati (1986), Government size and economic growth: a new framework and some evidence from cross section and time series data, American Economic Review, 76,191–203.
- There exists an optimal size of government.
 - If the govt size is small, and an increase in govt size will increase the rate of growth;
 - If govt size is too big, and an decrease in govt size will increase the rate of growth.
 - Barro, R. (1990), “Government Spending in a Simple Model of Endogenous Growth.” Journal of Political Economy 98, s103-125.

B. Taxes

- Increases in income taxes lead to permanent declines in the rate of economic growth.
 - Rebelo, Sergio, 1991, “Long Run Policy Analysis and Long Run Growth,” *Journal Political Economy*, 99:500-521.
- Tax structure matters for growth
 - Lee, Young, and Roger Gordon (2005) “Tax structure and economic growth,” *Journal of Public Economics*, 89, 1027-1043.
 - Based on cross-country data during 1970–1997, statutory corporate tax rates are significantly negatively correlated with cross-sectional differences in growth rates. The coefficient estimates suggest that a cut in the corporate tax rate by 10 percentage points will raise the annual growth rate by 1-2 percentage points.
 - Jens Arnold, 2008, Do Tax Structures Affect Aggregate Economic Growth? Empirical Evidence from a Panel of OECD Countries, OECD WP 643.
 - Based on panel data from 21 OECD countries, it is found that income taxes are generally associated with lower economic growth than taxes on consumption and property. More precisely, property taxes, and particularly recurrent taxes on immovable property, seem to be the most growth-friendly, followed by consumption taxes, and then by personal income taxes. Corporate income taxes appear to have the most negative effect on GDP per capita.

C. Government debt

- The economics profession seems to increasingly endorse the existence of a strongly negative nonlinear effect of public debt on economic growth.
- Reinhart and Rogoff (2010) "Growth in a Time of Debt," *American Economic Review*, 100(2), 573-78, May. NBER Working Paper No. 15639
 - A public debt-to-GDP ratio higher than 90% of GDP is associated with considerably lower economic performance in advanced and emerging economies alike.